

ANKIT SINGH

Full-Stack Platform Engineer | API & System Performance | AWS | Reliability, Observability & Incident Response

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Lead engineer with 6+ years of experience building and scaling customer-facing systems and APIs in fintech and cloud environments. Strong in system design, data modeling, reliability/observability, and performance optimization. Experienced in end-to-end architecture and production operations (monitoring, incident response, cost and performance tuning), and in working with product and ops to ship useful features.

Senior Software Engineer — Sequoia Jul 2024 - Present
Project: GenAI platform (Dify + RAG)

- **Built an internal GenAI agent platform (Dify + RAG)** that multiple teams reuse, with shared configs, runbooks, and guardrails, using **Python, FastAPI, and MongoDB**. Focused on the relevance of retrieved context and quality of responses for internal users.
- Refined **prompts and retrieval in Dify's Prompt IDE** to make answers more relevant and reduce bad responses.
- Led **technical decision-making and architectural tradeoffs**, aligned engineering work with business outcomes and system constraints.

Lead Engineer — LotusPay (acquired by Juspay) Feb 2020 - Jun 2024
Project: Developing India's leading NACH mandate and payments system

Public Platform & Developer Experience

- Owned **payment dashboards** using **Ruby on Rails and JavaScript** for **CRUD payment operations**, **simplifying workflows** for **support and ops teams**.
- Developed **targeted payment tracking pop-ups**, improving user clarity and **increasing payment completion from 69% to 90%** through enhanced visibility into transaction statuses.
- Optimized **AJAX requests** through **HTTP/2, enabled Keep-Alive, and batched requests**, helping **reduce page load time from 7s to 1s**, delivering a smoother user experience.

Reliability, Observability & Incident Response

- Engineered **scalable REST APIs (Rails, Redis, AWS EC2)**, improving **payment success rate from 50% to 90%**; implemented consistent error schemas, **pagination, and API versioning to reduce integration issues** and support escalations.
- **Improved API developer experience** by standardizing error responses and adding structured logs, cutting customer confusion and lifting CSAT by **25%**.
- Fixed critical endpoints with **idempotency keys** and **strict request validation** to prevent duplicate processing under retries.
- Built **event-driven, distributed processing using Sidekiq** workers with **Redis Streams** and **AWS SQS for queuing/retries**, **added idempotency + audit logs** to handle spikes safely and keep processing **reliable** under **failures**.
- **Improved performance with caching**, tuned **DB queries/indexes**, and **blue-green deployments**, and set up CloudWatch + SNS + Lambda alerts to **cut downtime by 95%** and keep incident resolution under **5 minutes**.
- Reworked our bank **payment data ingestion pipeline** by running **A/B tests on batch size, indexing, and concurrency settings** on **AWS RDS(MySQL)**, the winning setup cut **processing time by 15x** and made imports far more reliable under load.

Performance & Release Engineering

- Redesigned the **core MySQL schema on AWS RDS**, tuned **heavy queries (New Relic + EXPLAIN)**, and added **Redis + CDN caching**, cutting server **load by 50%** and **saving over \$10k annually**.
- Implemented **blue-green deployments** using **Route 53 + S3 + CloudFront**, enabling **zero-downtime rollouts** and reducing release-related disruptions by **30% during high-traffic hours**.
- **Led quarterly reliability and performance efforts** and **mentored engineers** through design and code reviews, improving code quality and release practices.

Skills

- Ruby on Rails, Python, NodeJS
 - MySQL, PostgreSQL, MongoDB
 - HTML, CSS, React, Socket Programming
 - Grafana, Prometheus, AppSignal, New Relic
 - Learning: TypeScript, GraphQL, Jest
- * Flask, FastAPI, Shell Scripting, REST APIs
 - * Redis, Sidekiq, Git, GitHub, OpenAI APIs
 - * [Certified Amazon Web Services Solution Architect](#)
 - * RSpec, PyTest, Microservices, Monorepo
 - * Docker

Tech Leadership Highlight

- Acted as a point of contact between product, support, and engineering, translating product priorities into architecture changes and capacity/reliability work.
- Published manuscript on [Object Detection and Segmentation](#) in an International Scientific Research Journal.
- **Collaborated with Support/Ops to turn incident and ticket trends into engineering work**, reducing recurring issues.

Education

B.Tech - Computer Science at Cambridge Institute of Technology (2016-2020)